

ZHENXIAO QI

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SUMMARY

Zhenxiao Qi is a final-year Ph.D. candidate in Computer Science from UC Riverside. He has 6+ years of research and software engineering experience on **Software** and **Distributed System Security**. He solves security problems with program analysis and automated reasoning, including **vulnerability assessment**, **automated test generation**, **cloud threat detection**, **kernel rootkit detection**, **memory forensics**, etc.

SKILLS

Skills: Cybersecurity, Program Analysis, SMT Solver, Distributed System, Cloud Threat Detection.

Languages: C/C++, Python, Java, Logic Programming, Z3.

Developer Tools: Google Cloud Platform, AWS, Kubernetes, Git, Docker, SQL.

Program Analysis: AFL/AFL++, Pytest Framework, Hybrid Fuzzing, Symbolic Execution.

EDUCATION

University of California, Riverside 2018 – 2024

Degree expected: Ph.D. in Computer Science

University of California, Riverside 2018 - 2022

Master of Science, Computer Science

Xidian University

Bachelor of Science, Information Security 2014-2018

PROFESSIONAL EXPERIENCE

Research Engineer Intern @ Google, Sunnyvale, CA Jun 2022 – Sep. 2022

- Implemented kernel rootkit detection by enforcing kernel data structure integrity.
- Built threat model and detection for Crypto mining malware on GCP VM.
- Designed and implemented algorithms in C++ to improve the efficiency of kernel memory region identification in large memory snapshots, leveraging Linux kernel page table features.

Research Engineer Intern @ Deepbits, Riverside, CA Oct. 2021 – Jun. 2022

- Proposed binary symbolic execution approach for bug detection and exploitation generation.
- Designed and implemented symbolic constraint collection for C/C++ programs, resulting in 10 times speedup on constraint collection.
- Designed optimistic constraint solving strategy to improve SMT solving throughput.

Research Engineer Intern @ Google, Remote Jun. 2021 – Sep. 2021

- Designed and implemented kernel memory analysis for GCP VMs using logic inference in Prolog, increasing threat detection coverage by 20%.

PUBLICATIONS

[USENIX Security'24] Zhenxiao Qi, Jie Hu, Zhaoqi Xiao, Heng Yin, [SymFit: Making the Common \(Concrete\) case faster for Binary-code Concolic Execution](#), to appear in the the 33rd USENIX Security Symposium.

[NDSS'22] Zhenxiao Qi, Yu Qu, Heng Yin, [LOGICMEM: Automatic Profile Generation for Binary-only Memory Forensics via Logic Inference](#), to appear in the Network and Distributed System Security Symposium, February 2022.

[NDSS'21] Zhenxiao Qi, Qian Feng, Yueqiang Cheng, Mengjia Yan, Peng Li, Heng Yin, and Tao Wei, [SpecTaint: Speculative Taint Analysis for Discovering Spectre Gadgets](#), to appear in the Network and Distributed System Security Symposium, February 2021

[RAID'19] Ali Davanian, Zhenxiao Qi, Yu Qu, and Heng Yin, [DECAF++: Elastic Whole-System Dynamic Taint Analysis](#), in the 22nd International Symposium on Research in Attacks, Intrusions and Defenses, September 2019

RESEARCH PROJECTS

Elastic binary-code concolic execution (published in USENIX Security '24)

- Developed binary symbolic execution tool in C++ for bug detection and exploitation generation.
- Proposed an elastic analysis scheme that boost up existing tool's performance by 10 times.
- Improved code coverage growth and constraint solving efficiency in hybrid fuzzing experiments.

Binary-only profile generation for kernel integrity analysis (NDSS '22)

- Proposed and implemented a novel solution in Prolog to generate profiles for binary-only memory forensics using a logic inference approach.
- Recover kernel data structures from memory snapshot with 100% precision and 95% recall on average.

SpecTaint: Speculative Taint Analysis for Discovering Spectre Gadgets (NDSS '21)

- Developed a novel Spectre gadget detection tool in C++ by enabling dynamic taint analysis on simulated speculative execution paths.
- Improved the detection accuracy and efficiency by 73% compared to existing tools.
- Detected 11 new Spectre gadgets from real-world programs such as Brotli and Caffee.

Cloud native dynamic container monitoring

- Developed a runtime container monitor service in C without running an agent inside the container.
- Automatically collect activities of running processes inside the AWS ECS, such as network connection, opened files, process credentials and permissions, etc.
- Performed large-scale analysis of existing docker images hosted on DockerHub.

Memory snapshot analysis for kernel rootkit and threat detection

- Developed an API-based service in Python for memory snapshot analysis.
- Developed a differential testing approach to detecting hidden kernel rootkits and threats.